

Can a Hopfield Network Learn Like a Living System?

A NeurIPS 2023 position paper distills three biological adaptation mechanisms and proposes concrete ways to fold them into the Hopfield-network self-optimization model.

Modern deep learning is powerful, but it learns in a way no organism does. It needs enormous datasets, supervised or reward-based training signals, and a great deal of energy, and when it learns a new task in sequence it tends to overwrite the old one, a failure mode known as catastrophic forgetting. Living systems face the same world with almost the opposite budget: they learn and adapt from sparse data, spend little energy, and rarely need many training iterations to make a useful association.

That gap is the starting point for this NeurIPS 2023 position paper by Aisha Belhadi at the Okinawa Institute of Science and Technology. Rather than scale a network up, it asks a more biological question: which mechanisms let living systems stay adaptive, and can we translate them into an associative-memory model directly? The paper distills three such mechanisms and proposes concrete, low-cost ways to build each one into the Hopfield-network self-optimization model.

The move is deliberately economical. Instead of adding parameters, the paper argues for raising the network's degrees of freedom by letting components it already has do more work, exactly the trick biology uses. The figure below is the map for the whole argument: three mechanisms of biological adaptation on the left, and three matching places to intervene in the model shown on the right side.

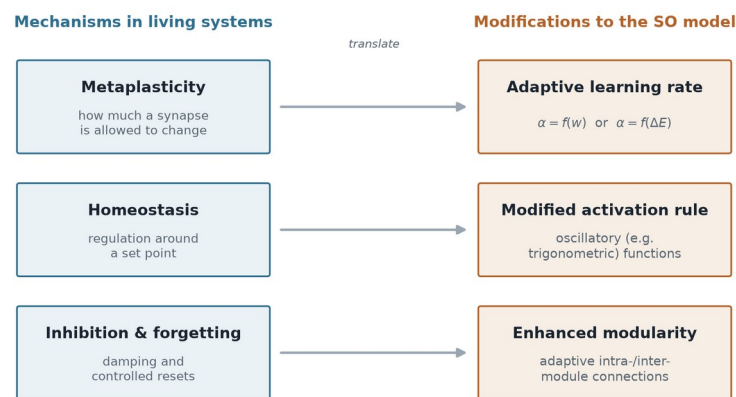


Figure 1. The paper's core proposal, mapping three mechanisms of biological adaptation onto three concrete modifications of the self-optimization (SO) model. This is a schematic of relationships the paper describes in prose; the paper itself contains no figures.

Paper: <https://openreview.net/forum?id=WjHYgEfXiv>

The tension the paper is really about

To act in a changing world, any system has to integrate new information while staying stable enough to keep what it already knows. Push too hard toward novelty and learning never settles; lean too hard on the past and the system locks in stale patterns. Biology holds that balance through an interplay of mechanisms, and the paper reads the same robustness-versus-flexibility problem that machine learning keeps running into. Its wager is that the mechanisms biology uses to manage the balance are the missing ingredients, not more scale.

The chosen substrate is the self-optimization (SO) model, a Hopfield network with Hebbian weight updates and periodic resets of the node states. Left alone, a standard Hopfield network is a limited associative memory that reliably stores only about $0.14N$ patterns for N nodes. The SO model is more interesting because the resets let it build a memory of the attractors it has visited and gradually enlarge the basin around the global energy minimum, behavior that already rhymes with biological attractor dynamics. That makes it a natural place to add more biology.

Lever one: make the learning rate adaptive

The first and most developed proposal targets metaplasticity, the biological property that governs how much a synapse is allowed to change. In a plain Hebbian rule the learning rate is a fixed constant the user picks. The paper makes it a function instead, and it describes two flavors. In local metaplasticity the rate follows each connection's own weight, so the update to a weight becomes a function of that weight times the two node states, potentiating co-active pairs quickly and depressing opposing ones slowly. In network-level metaplasticity the rate follows the change in the network's energy, which behaves like gradient descent on the energy landscape.

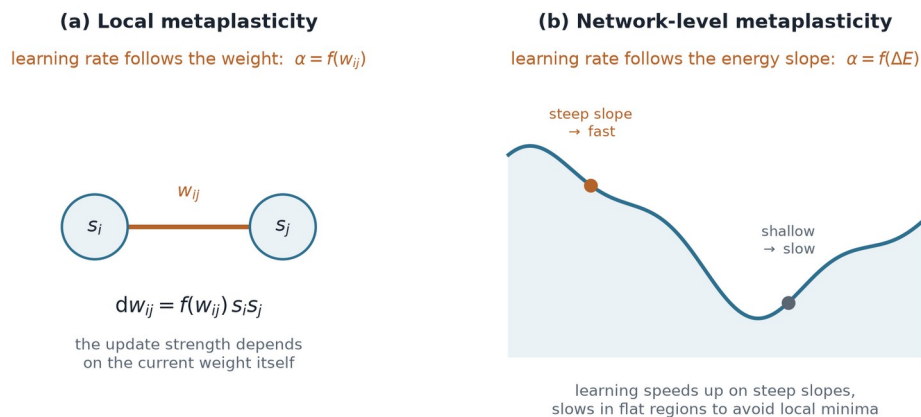


Figure 2. Two forms of metaplasticity. (a) Local: the learning rate follows each connection's own weight, so the Hebbian update becomes weight-dependent. (b) Network-level: the learning rate follows the change in network energy, speeding up on steep slopes and slowing in flat regions to help escape local minima.

Both versions cost nothing extra in parameters. They reuse quantities the network already computes, the weight and the energy, and turn them into a self-regulating dial on plasticity. The energy-driven form is the more suggestive of the two: learning accelerates on the steep parts of

the landscape and eases off where it flattens, which is a plausible way to avoid getting stuck in shallow local minima.

Levers two and three: activation and modularity

Homeostasis, the regulation of a system around a set point, motivates the second intervention: swap the usual activation rule for an oscillatory one, such as a trigonometric function, so the network's dynamics carry a built-in rhythm around a baseline rather than settling flat. The third intervention targets structure. Modular architecture with adaptive intra- and inter-module connections, together with inhibition between modules, supports temporal integration, degeneracy, and continuous learning, and it echoes the biological idea of synaptic tagging and capture, where a module that keeps strengthening its internal links earns an automatic boost while inhibition keeps overall energy in check.

Why constraints are the point, not the price

The most interesting claim in the paper is conceptual. It is tempting to read biological constraints as the opposite of flexibility, as rules that hem a system in. The paper argues the reverse: constraints and degrees of freedom are complementary. The architecture supplies a constraining frame, and within that frame added degrees of freedom can open up genuinely new adaptive behavior. Remove the frame and the same added freedom has nothing to shape it, so it collapses into uninformative noise.

Constraints give shape to added degrees of freedom

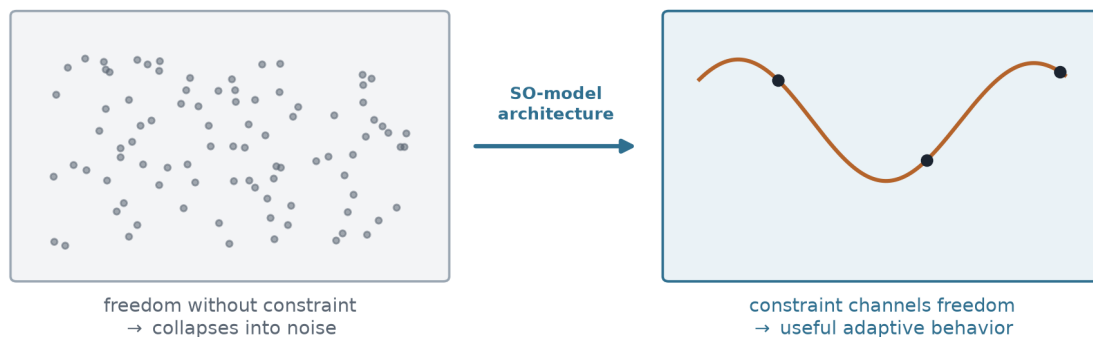


Figure 3. The unifying idea. Architectural constraints are what let added degrees of freedom become useful, structured adaptation rather than collapsing into noise.

What this is, and what it is not

It is worth being clear about the paper's status. This is a proposal, not an empirical study: there is no dataset, no benchmark, and no trained model. Its result is a coherent design framework, mapping metaplasticity, homeostasis, and inhibition onto an adaptive learning rate, a modified activation rule, and adaptive modularity, and it grounds those ideas in prior

computational work on the SO model, including a model of the *C. elegans* connectome and SO models that solve combinatorial satisfiability problems. Each of the proposed functions still has to be defined algorithmically and then tested in careful follow-up experiments.

Read that way, the contribution is a research agenda with an unusually clear organizing principle. If associative-memory models are to learn adaptively from sparse data the way living systems do, the paper's bet is that the path runs through making plasticity self-regulating and letting structure, not scale, do the constraining. Whether the specific functions pan out is exactly the empirical question the paper hands to whoever picks it up next.

For practitioners, the concrete takeaway is small but pointed. Before reaching for more data or more parameters, it may be worth asking whether some quantity the system already tracks, a weight or an energy value, could be made to regulate learning on its own, and whether structure rather than scale could do the work of holding the model steady.